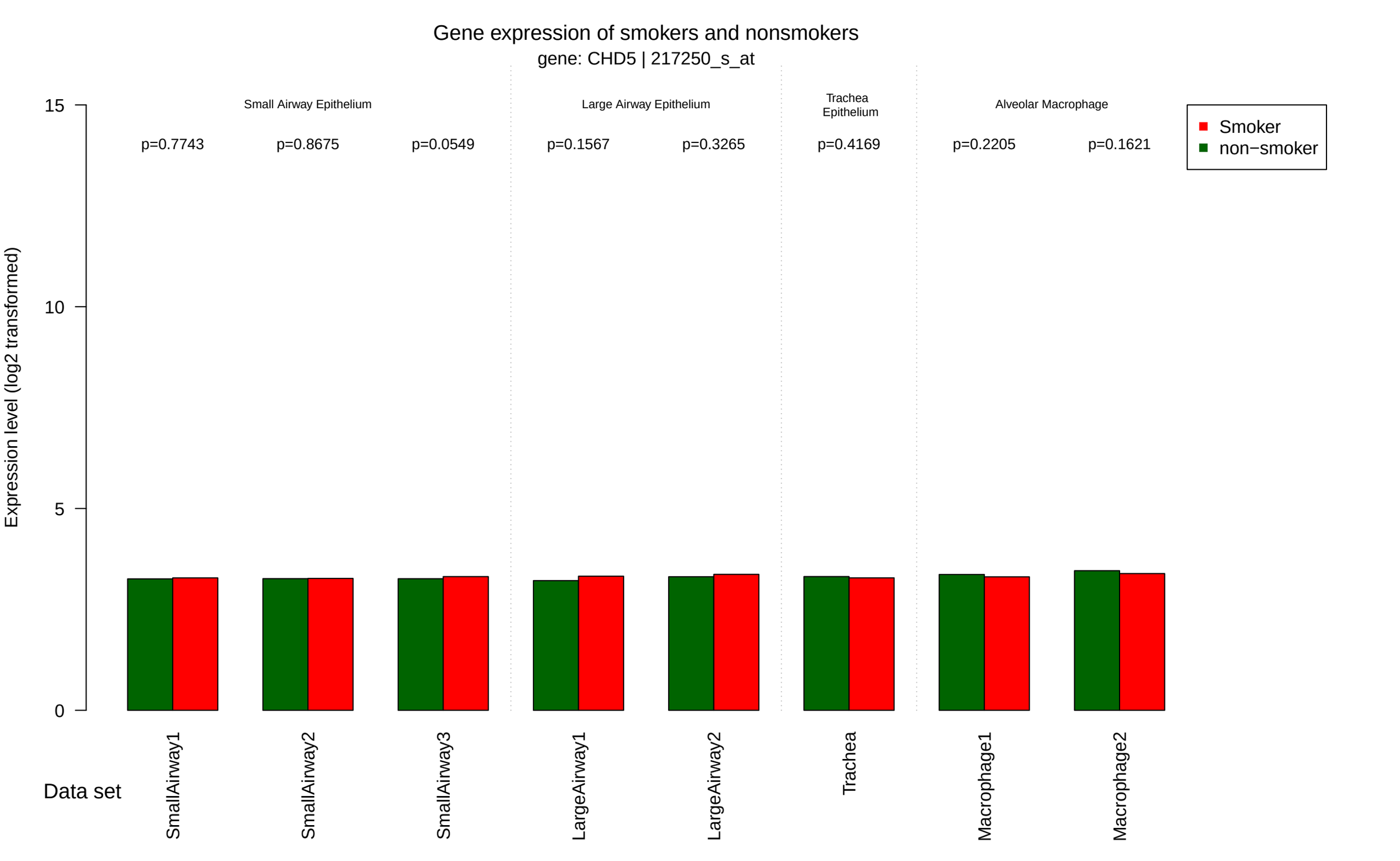
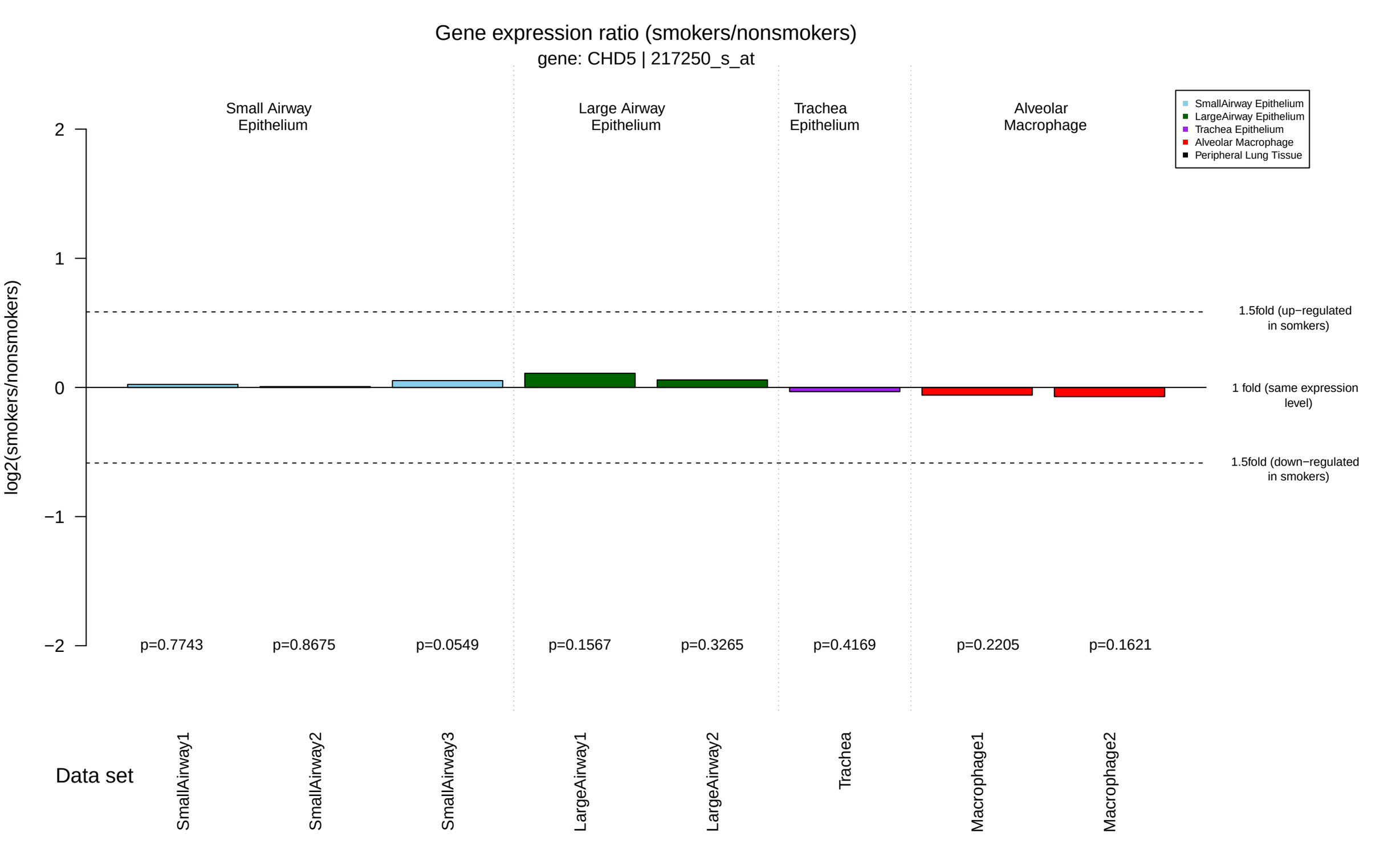
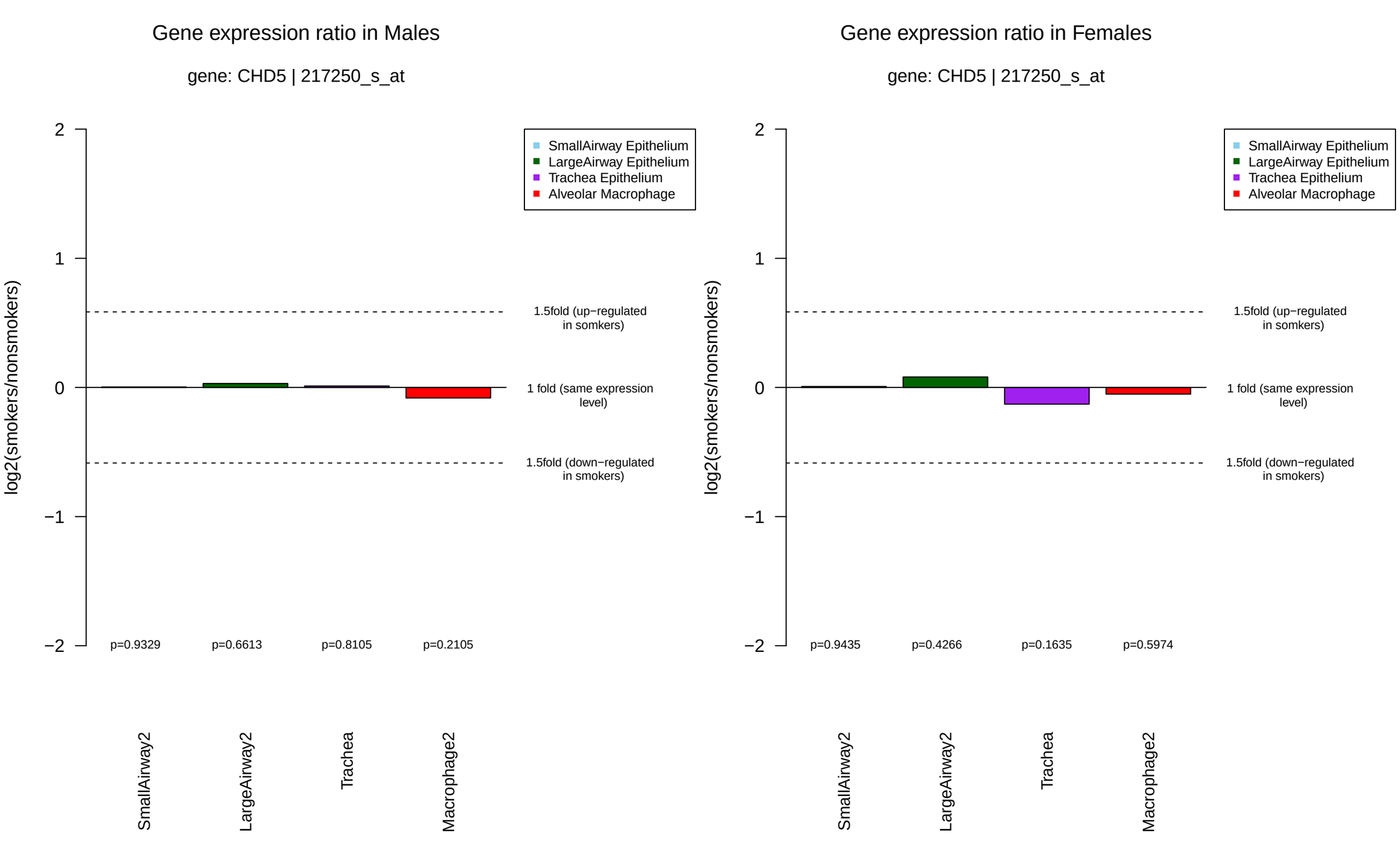
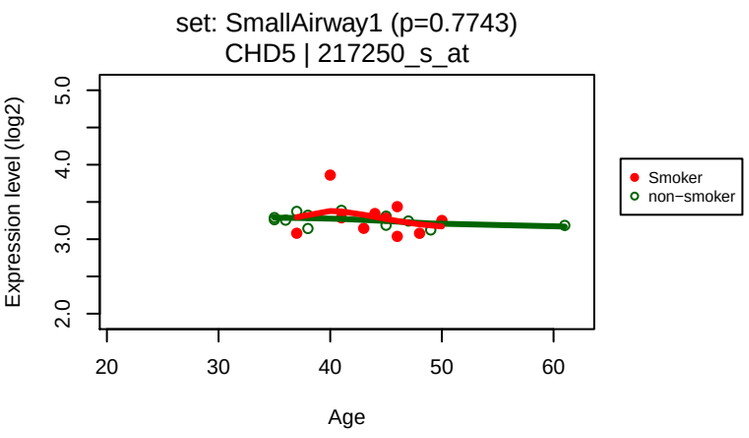
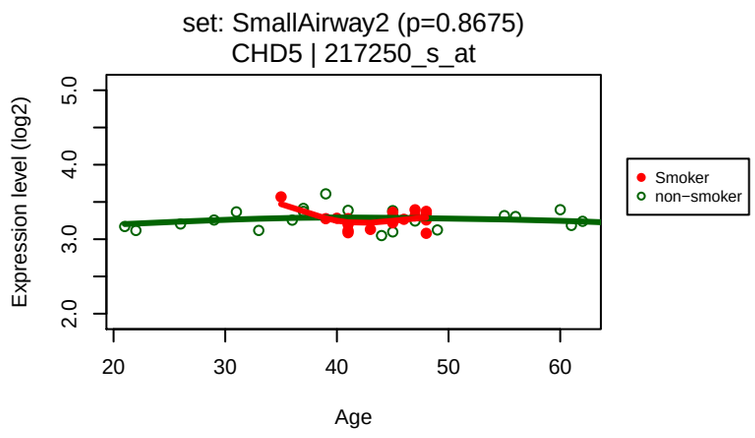
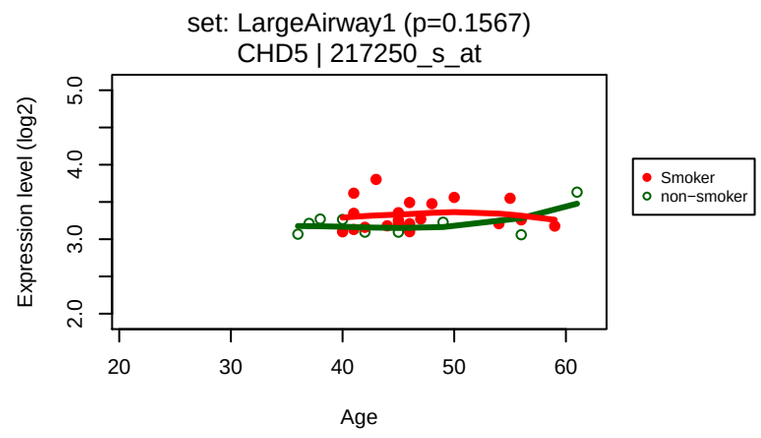

Gene expression of smokers and nonsmokers
gene: CHD5 | 217250_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7743
p=0.8675
p=0.0549
p=0.1567
p=0.3265
p=0.4169
p=0.2205
p=0.1621
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

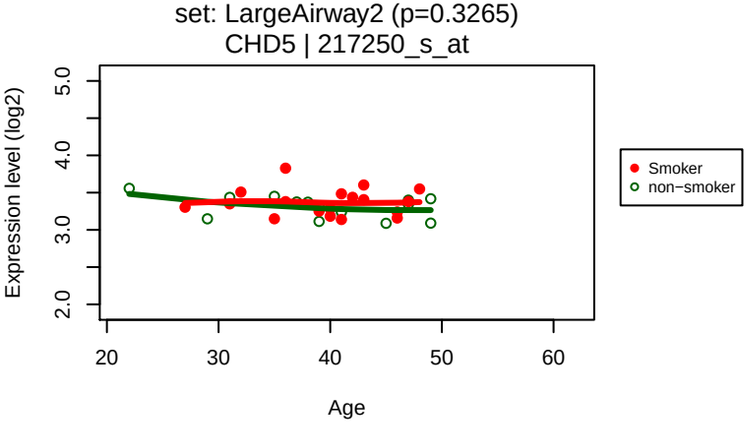

Gene expression ratio (smokers/nonsmokers)
gene: CHD5 | 217250_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7743
p=0.8675
p=0.0549
p=0.1567
p=0.3265
p=0.4169
p=0.2205
p=0.1621
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

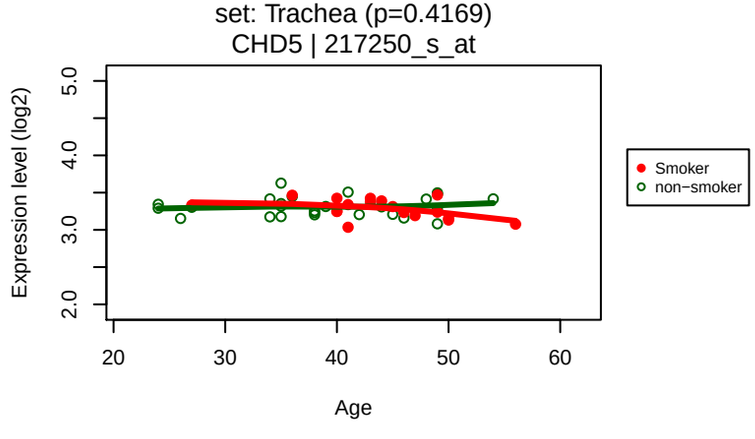

Gene expression ratio in Males
gene: CHD5 | 217250_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9329
p=0.6613
p=0.8105
p=0.2105
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHD5 | 217250_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9435
p=0.4266
p=0.1635
p=0.5974
SmallAirway2
LargeAirway2
Trachea
Macrophage2

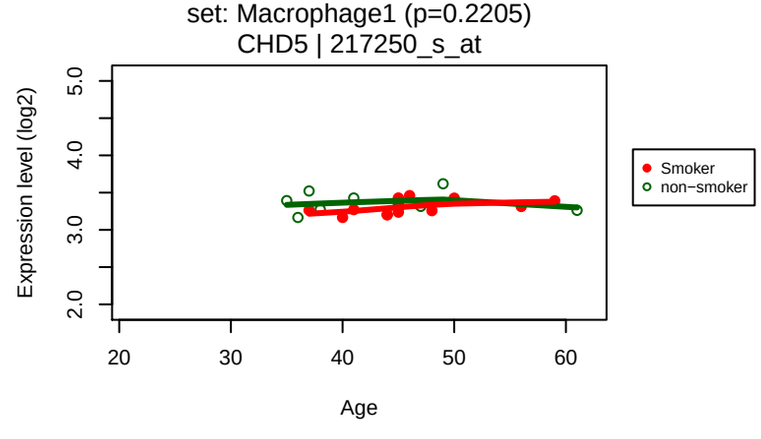

set: SmallAirway1 (p=0.7743)
CHD5 | 217250_s_at
Expression level (log2)
Age
Smoker
non−smoker

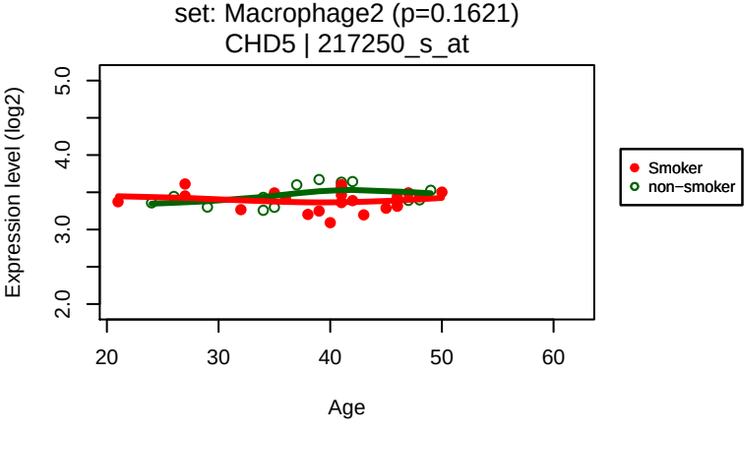

set: SmallAirway2 (p=0.8675)
CHD5 | 217250_s_at
Expression level (log2)
Age
Smoker
non−smoker

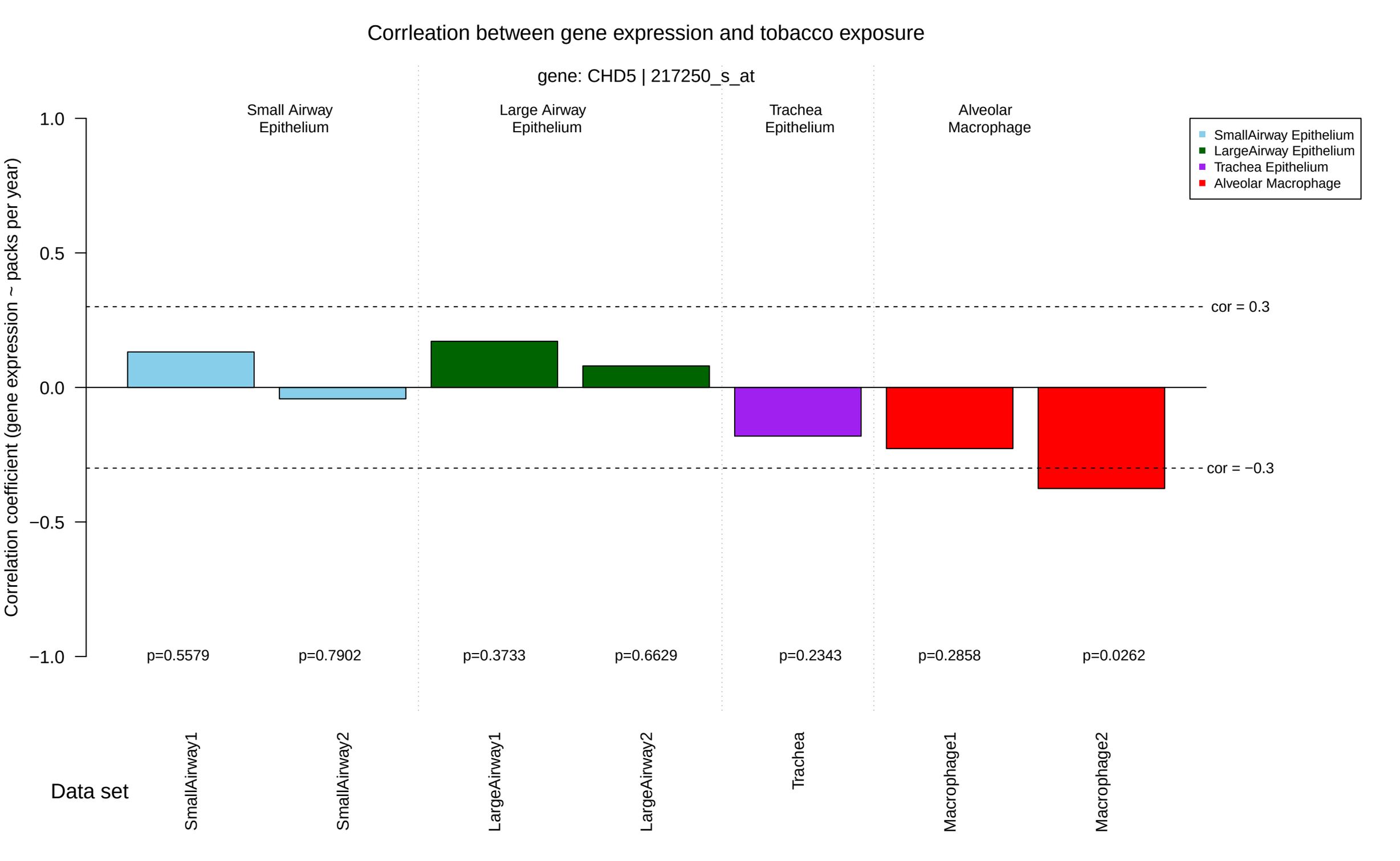

set: LargeAirway1 (p=0.1567)
CHD5 | 217250_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3265)
CHD5 | 217250_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4169)
CHD5 | 217250_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2205)
CHD5 | 217250_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1621)
CHD5 | 217250_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHD5 | 217250_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5579
p=0.7902
p=0.3733
p=0.6629
p=0.2343
p=0.2858
p=0.0262
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2